Dear Apple hiring team,

I am writing to apply for the AR/VR Software Development Engineer position within Vision Products Software at Apple. My background is research in theoretical physics at Oxford, particularly gauge-theory gravity, a theory that uses coordinate-free geometric algebra, whose mathematics has applications in computer vision. Primarily in creating a mathematical framework that prevents computations that err as a result of coordinate-singularities and gimbal-lock, and can only produce quantities that transform covariantly, and thus seamlessly permit a moving and rotating camera.

You will find few candidates who both have a technical mathematical knowledge of intrinsic-geometric 3D pose estimation, camera recalibration, local rotation fields, and the ability to program them into software a camera or computer can use.

My skills in software development come from MIT, where I took classes in Software Construction, Algorithms, and Computer Systems. I applied these algorithmic principles during my undergraduate research, where I built a computer vision pipeline to track silicone droplets. The system required image processing, dynamic thresholding, edge detection, and real-time path estimation.

The core computational requirements of Vision Products Software—real-time spatial tracking, local geometric transformations, sensor fusion, and high-performance algorithms—align with my background. I offer a combination of expertise in geometric algebra, gauge-invariant theory, and practical programming implementation.

I welcome the opportunity to work for the Vision Products team.

Sincerely,

Isabelle Rose